

Analysis of Variance in Multiple Regression

The partition of the total sum of squares into its components, the regression and error sums of squares, plays an important role. An **analysis of variance** can be conducted to shed light on the quality of the regression equation. A useful hypothesis that determines if a significant amount of variation is explained by the model is

$$H_0: \beta_1 = \beta_2 = \beta_3 = \cdots = \beta_k = 0.$$

The analysis of variance involves an F -test via a table given as follows:

Source	Sum of Squares	Degrees of Freedom	Mean Squares	F
Regression	SSR	k	$MSR = \frac{SSR}{k}$	$f = \frac{MSR}{MSE}$
Error	SSE	$n - (k + 1)$	$MSE = \frac{SSE}{n - (k + 1)}$	
Total	SST	$n - 1$		

This test is an **upper-tailed test**. Rejection of H_0 implies that the **regression equation differs from a constant**. That is, at least one regressor variable is important. More discussion of the use of analysis of variance appears in subsequent sections.

Further utility of the mean square error (or residual mean square) lies in its use in hypothesis testing and confidence interval estimation, which is discussed in Section 12.5. In addition, the mean square error plays an important role in situations where the scientist is searching for the best from a set of competing models. Many model-building criteria involve the statistic s^2 . Criteria for comparing competing models are discussed in Section 12.11.

12.5 Inferences in Multiple Linear Regression

A knowledge of the distributions of the individual coefficient estimators enables the experimenter to construct confidence intervals for the coefficients and to test hypotheses about them. Recall from Section 12.4 that the b_j ($j = 0, 1, 2, \dots, k$) are normally distributed with mean β_j and variance $c_{jj}\sigma^2$. Thus, we can use the statistic

$$t = \frac{b_j - \beta_{j0}}{s\sqrt{c_{jj}}}$$

with $n - k - 1$ degrees of freedom to test hypotheses and construct confidence intervals on β_j . For example, if we wish to test

$$\begin{aligned} H_0: \beta_j &= \beta_{j0}, \\ H_1: \beta_j &\neq \beta_{j0}, \end{aligned}$$

we compute the above t -statistic and do not reject H_0 if $-t_{\alpha/2} < t < t_{\alpha/2}$, where $t_{\alpha/2}$ has $n - k - 1$ degrees of freedom.

Example 12.5: For the model of Example 12.4, test the hypothesis that $\beta_2 = -2.5$ at the 0.05 level of significance against the alternative that $\beta_2 > -2.5$.

Solution:

$$H_0: \beta_2 = -2.5,$$

$$H_1: \beta_2 > -2.5.$$

Computations:

$$t = \frac{b_2 - \beta_{20}}{s\sqrt{c_{22}}} = \frac{-1.8616 + 2.5}{2.073\sqrt{0.0166}} = 2.390,$$

$$P = P(T > 2.390) = 0.04.$$

Decision: Reject H_0 and conclude that $\beta_2 > -2.5$. ┐

Individual t -Tests for Variable Screening

The t -test most often used in multiple regression is the one that tests the importance of individual coefficients (i.e., $H_0: \beta_j = 0$ against the alternative $H_1: \beta_j \neq 0$). These tests often contribute to what is termed **variable screening**, where the analyst attempts to arrive at the most useful model (i.e., the choice of which regressors to use). It should be emphasized here that if a coefficient is found insignificant (i.e., the hypothesis $H_0: \beta_j = 0$ **is not rejected**), the conclusion drawn is that the **variable** is insignificant (i.e., explains an insignificant amount of variation in y), **in the presence of the other regressors in the model**. This point will be reaffirmed in a future discussion.

Inferences on Mean Response and Prediction

One of the most useful inferences that can be made regarding the quality of the predicted response y_0 corresponding to the values $x_{10}, x_{20}, \dots, x_{k0}$ is the confidence interval on the mean response $\mu_{Y|x_{10}, x_{20}, \dots, x_{k0}}$. We are interested in constructing a confidence interval on the mean response for the set of conditions given by

$$\mathbf{x}'_0 = [1, x_{10}, x_{20}, \dots, x_{k0}].$$

We augment the conditions on the x 's by the number 1 in order to facilitate the matrix notation. Normality in the ϵ_i produces normality in the b_j and the mean and variance are still the same as indicated in Section 12.4. So is the covariance between b_i and b_j , for $i \neq j$. Hence,

$$\hat{y} = b_0 + \sum_{j=1}^k b_j x_{j0}$$

is likewise normally distributed and is, in fact, an unbiased estimator for the **mean response** on which we are attempting to attach a confidence interval. The variance of $\hat{y}_0$, written in matrix notation simply as a function of σ^2 , $(\mathbf{X}'\mathbf{X})^{-1}$, and the condition vector $\mathbf{x}'_0$, is

$$\sigma_{\hat{y}_0}^2 = \sigma^2 \mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0.$$

If this expression is expanded for a given case, say $k = 2$, it is readily seen that it appropriately accounts for the variance of the b_j and the covariance of b_i and b_j , for $i \neq j$. After σ^2 is replaced by s^2 as given by Theorem 12.1, the $100(1 - \alpha)\%$ confidence interval on $\mu_{Y|x_{10}, x_{20}, \dots, x_{k0}}$ can be constructed from the statistic

$$T = \frac{\hat{y}_0 - \mu_{Y|x_{10}, x_{20}, \dots, x_{k0}}}{s \sqrt{\mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0}},$$

which has a t -distribution with $n - k - 1$ degrees of freedom.

Confidence Interval for $\mu_{Y|x_{10}, x_{20}, \dots, x_{k0}}$ A $100(1 - \alpha)\%$ confidence interval for the **mean response** $\mu_{Y|x_{10}, x_{20}, \dots, x_{k0}}$ is

$$\hat{y}_0 - t_{\alpha/2} s \sqrt{\mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0} < \mu_{Y|x_{10}, x_{20}, \dots, x_{k0}} < \hat{y}_0 + t_{\alpha/2} s \sqrt{\mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0},$$

where $t_{\alpha/2}$ is a value of the t -distribution with $n - k - 1$ degrees of freedom.

The quantity $s \sqrt{\mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0}$ is often called the **standard error of prediction** and appears on the printout of many regression computer packages.

Example 12.6: Using the data of Example 12.4, construct a 95% confidence interval for the mean response when $x_1 = 3\%$, $x_2 = 8\%$, and $x_3 = 9\%$.

Solution: From the regression equation of Example 12.4, the estimated percent survival when $x_1 = 3\%$, $x_2 = 8\%$, and $x_3 = 9\%$ is

$$\hat{y} = 39.1574 + (1.0161)(3) - (1.8616)(8) - (0.3433)(9) = 24.2232.$$

Next, we find that

$$\begin{aligned} \mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0 &= [1, 3, 8, 9] \begin{bmatrix} 8.0648 & -0.0826 & -0.0942 & -0.7905 \\ -0.0826 & 0.0085 & 0.0017 & 0.0037 \\ -0.0942 & 0.0017 & 0.0166 & -0.0021 \\ -0.7905 & 0.0037 & -0.0021 & 0.0886 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 8 \\ 9 \end{bmatrix} \\ &= 0.1267. \end{aligned}$$

Using the mean square error, $s^2 = 4.298$ or $s = 2.073$, and Table A.4, we see that $t_{0.025} = 2.262$ for 9 degrees of freedom. Therefore, a 95% confidence interval for the mean percent survival for $x_1 = 3\%$, $x_2 = 8\%$, and $x_3 = 9\%$ is given by

$$\begin{aligned} 24.2232 - (2.262)(2.073)\sqrt{0.1267} &< \mu_{Y|3,8,9} \\ &< 24.2232 + (2.262)(2.073)\sqrt{0.1267}, \end{aligned}$$

or simply $22.5541 < \mu_{Y|3,8,9} < 25.8923$. ▮

As in the case of simple linear regression, we need to make a clear distinction between the confidence interval on a mean response and the prediction interval on an *observed response*. The latter provides a bound within which we can say with a preselected degree of certainty that a new observed response will fall.

A prediction interval for a single predicted response y_0 is once again established by considering the difference $\hat{y}_0 - y_0$. The sampling distribution can be shown to be normal with mean

$$\mu_{\hat{y}_0 - y_0} = 0$$

and variance

$$\sigma_{\hat{y}_0 - y_0}^2 = \sigma^2[1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0].$$

Thus, a $100(1 - \alpha)\%$ prediction interval for a single prediction value y_0 can be constructed from the statistic

$$T = \frac{\hat{y}_0 - y_0}{s\sqrt{1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}},$$

which has a t -distribution with $n - k - 1$ degrees of freedom.

Prediction Interval for y_0 A $100(1 - \alpha)\%$ prediction interval for a **single response** y_0 is given by

$$\hat{y}_0 - t_{\alpha/2}s\sqrt{1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0} < y_0 < \hat{y}_0 + t_{\alpha/2}s\sqrt{1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0},$$

where $t_{\alpha/2}$ is a value of the t -distribution with $n - k - 1$ degrees of freedom.

Example 12.7: Using the data of Example 12.4, construct a 95% prediction interval for an individual percent survival response when $x_1 = 3\%$, $x_2 = 8\%$, and $x_3 = 9\%$.

Solution: Referring to the results of Example 12.6, we find that the 95% prediction interval for the response y_0 , when $x_1 = 3\%$, $x_2 = 8\%$, and $x_3 = 9\%$, is

$$24.2232 - (2.262)(2.073)\sqrt{1.1267} < y_0 < 24.2232 + (2.262)(2.073)\sqrt{1.1267},$$

which reduces to $19.2459 < y_0 < 29.2005$. Notice, as expected, that the prediction interval is considerably wider than the confidence interval for mean percent survival found in Example 12.6. ■

Annotated Printout for Data of Example 12.4

Figure 12.1 shows an annotated computer printout for a multiple linear regression fit to the data of Example 12.4. The package used is *SAS*.

Note the model parameter estimates, the standard errors, and the t -statistics shown in the output. The standard errors are computed from square roots of diagonal elements of $(\mathbf{X}'\mathbf{X})^{-1}s^2$. In this illustration, the variable x_3 is insignificant in the presence of x_1 and x_2 based on the t -test and the corresponding P -value of 0.5916. The terms CLM and CLI are confidence intervals on mean response and prediction limits on an individual observation, respectively. The f -test in the analysis of variance indicates that a significant amount of variability is explained. As an example of the interpretation of CLM and CLI, consider observation 10. With an observation of 25.2000 and a predicted value of 26.0676, we are 95% confident that the mean response is between 24.5024 and 27.6329, and a new observation will fall between 21.1238 and 31.0114 with probability 0.95. The R^2 value of 0.9117 implies that the model explains 91.17% of the variability in the response. More discussion about R^2 appears in Section 12.6.

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	399.45437	133.15146	30.98	<.0001
Error	9	38.67640	4.29738		
Corrected Total	12	438.13077			
Root MSE	2.07301	R-Square	0.9117		
Dependent Mean	29.03846	Adj R-Sq	0.8823		
Coeff Var	7.13885				

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	39.15735	5.88706	6.65	<.0001
x1	1	1.01610	0.19090	5.32	0.0005
x2	1	-1.86165	0.26733	-6.96	<.0001
x3	1	-0.34326	0.61705	-0.56	0.5916

Obs	Variable	Dependent Value	Predicted Mean	Std Error Predict	95% CL Mean	95% CL Predict	Residual	
1	25.5000	27.3514	1.4152	24.1500	30.5528	21.6734	33.0294	-1.8514
2	31.2000	32.2623	0.7846	30.4875	34.0371	27.2482	37.2764	-1.0623
3	25.9000	27.3495	1.3588	24.2757	30.4234	21.7425	32.9566	-1.4495
4	38.4000	38.3096	1.2818	35.4099	41.2093	32.7960	43.8232	0.0904
5	18.4000	15.5447	1.5789	11.9730	19.1165	9.6499	21.4395	2.8553
6	26.7000	26.1081	1.0358	23.7649	28.4512	20.8658	31.3503	0.5919
7	26.4000	28.2532	0.8094	26.4222	30.0841	23.2189	33.2874	-1.8532
8	25.9000	26.2219	0.9732	24.0204	28.4233	21.0414	31.4023	-0.3219
9	32.0000	32.0882	0.7828	30.3175	33.8589	27.0755	37.1008	-0.0882
10	25.2000	26.0676	0.6919	24.5024	27.6329	21.1238	31.0114	-0.8676
11	39.7000	37.2524	1.3070	34.2957	40.2090	31.7086	42.7961	2.4476
12	35.7000	32.4879	1.4648	29.1743	35.8015	26.7459	38.2300	3.2121
13	26.5000	28.2032	0.9841	25.9771	30.4294	23.0122	33.3943	-1.7032

Figure 12.1: SAS printout for data in Example 12.4.

More on Analysis of Variance in Multiple Regression (Optional)

In Section 12.4, we discussed briefly the partition of the total sum of squares $\sum_{i=1}^n (y_i - \bar{y})^2$ into its two components, the regression model and error sums of squares (illustrated in Figure 12.1). The analysis of variance leads to a test of

$$H_0: \beta_1 = \beta_2 = \beta_3 = \cdots = \beta_k = 0.$$

Rejection of the null hypothesis has an important interpretation for the scientist or engineer. (For those who are interested in more extensive treatment of the subject using matrices, it is useful to discuss the development of these sums of squares used in ANOVA.)

First, recall in Section 12.3, $\mathbf{b}$, the vector of least squares estimators, is given by

$$\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

A partition of the **uncorrected sum of squares**

$$\mathbf{y}'\mathbf{y} = \sum_{i=1}^n y_i^2$$

into two components is given by

$$\begin{aligned}\mathbf{y}'\mathbf{y} &= \mathbf{b}'\mathbf{X}'\mathbf{y} + (\mathbf{y}'\mathbf{y} - \mathbf{b}'\mathbf{X}'\mathbf{y}) \\ &= \mathbf{y}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} + [\mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}].\end{aligned}$$

The second term (in brackets) on the right-hand side is simply the error sum of squares $\sum_{i=1}^n (y_i - \hat{y}_i)^2$. The reader should see that an alternative expression for the error sum of squares is

$$SSE = \mathbf{y}'[\mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']\mathbf{y}.$$

The term $\mathbf{y}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ is called the **regression sum of squares**. However, it is not the expression $\sum_{i=1}^n (\hat{y}_i - \bar{y})^2$ used for testing the “importance” of the terms $b_1, b_2, \dots, b_k$ but, rather,

$$\mathbf{y}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \sum_{i=1}^n \hat{y}_i^2,$$

which is a regression sum of squares uncorrected for the mean. As such, it would only be used in testing if the regression equation differs significantly from zero, that is,

$$H_0: \beta_0 = \beta_1 = \beta_2 = \dots = \beta_k = 0.$$

In general, this is not as important as testing

$$H_0: \beta_1 = \beta_2 = \dots = \beta_k = 0,$$

since the latter states that the mean response is a constant, not necessarily zero.

Degrees of Freedom

Thus, the partition of sums of squares and degrees of freedom reduces to

Source	Sum of Squares	d.f.
Regression	$\sum_{i=1}^n \hat{y}_i^2 = \mathbf{y}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$	$k + 1$
Error	$\sum_{i=1}^n (y_i - \hat{y}_i)^2 = \mathbf{y}'[\mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']\mathbf{y}$	$n - (k + 1)$
Total	$\sum_{i=1}^n y_i^2 = \mathbf{y}'\mathbf{y}$	n

Hypothesis of Interest

Now, of course, the hypotheses of interest for an ANOVA must eliminate the role of the intercept described previously. Strictly speaking, if $H_0: \beta_1 = \beta_2 = \cdots = \beta_k = 0$, then the estimated regression line is merely $\hat{y}_i = \bar{y}$. As a result, we are actually seeking evidence that the regression equation “varies from a constant.” Thus, the total and regression sums of squares must be corrected for the mean. As a result, we have

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

In matrix notation this is simply

$$\mathbf{y}'[\mathbf{I}_n - \mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}']\mathbf{y} = \mathbf{y}'[\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - \mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}']\mathbf{y} + \mathbf{y}'[\mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']\mathbf{y}.$$

In this expression, $\mathbf{1}$ is a vector of n ones. As a result, we are merely subtracting

$$\mathbf{y}'\mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}'\mathbf{y} = \frac{1}{n} \left(\sum_{i=1}^n y_i \right)^2$$

from $\mathbf{y}'\mathbf{y}$ and from $\mathbf{y}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ (i.e., correcting the total and regression sums of squares for the mean).

Finally, the appropriate partitioning of sums of squares with degrees of freedom is as follows:

Source	Sum of Squares	d.f.
Regression	$\sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \mathbf{y}'[\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - \mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}']\mathbf{y}$	k
Error	$\sum_{i=1}^n (y_i - \hat{y}_i)^2 = \mathbf{y}'[\mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']\mathbf{y}$	$n - (k + 1)$
Total	$\sum_{i=1}^n (y_i - \bar{y})^2 = \mathbf{y}'[\mathbf{I}_n - \mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}']\mathbf{y}$	$n - 1$

This is the ANOVA table that appears in the computer printout of Figure 12.1. The expression $\mathbf{y}'[\mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}']\mathbf{y}$ is often called the **regression sum of squares associated with the mean**, and 1 degree of freedom is allocated to it.

Exercises

12.17 For the data of Exercise 12.2 on page 450, estimate σ^2 .

12.18 For the data of Exercise 12.1 on page 450, estimate σ^2 .

12.19 For the data of Exercise 12.5 on page 450, estimate σ^2 .

12.20 Obtain estimates of the variances and the co-

variance of the estimators b_1 and b_2 of Exercise 12.2 on page 450.

12.21 Referring to Exercise 12.5 on page 450, find the estimate of

- (a) $\sigma_{b_2}^2$;
- (b) $\text{Cov}(b_1, b_4)$.

12.22 For the model of Exercise 12.7 on page 451,

test the hypothesis that $\beta_2 = 0$ at the 0.05 level of significance against the alternative that $\beta_2 \neq 0$.

12.23 For the model of Exercise 12.2 on page 450, test the hypothesis that $\beta_1 = 0$ at the 0.05 level of significance against the alternative that $\beta_1 \neq 0$.

12.24 For the model of Exercise 12.1 on page 450, test the hypotheses that $\beta_1 = 2$ against the alternative that $\beta_1 \neq 2$. Use a P -value in your conclusion.

12.25 Using the data of Exercise 12.2 on page 450 and the estimate of σ^2 from Exercise 12.17, compute 95% confidence intervals for the predicted response and the mean response when $x_1 = 900$ and $x_2 = 1.00$.

12.26 For Exercise 12.8 on page 451, construct a 90% confidence interval for the mean compressive strength when the concentration is $x = 19.5$ and a quadratic model is used.

12.27 Using the data of Exercise 12.5 on page 450 and the estimate of σ^2 from Exercise 12.19, compute 95% confidence intervals for the predicted response and the mean response when $x_1 = 75$, $x_2 = 24$, $x_3 = 90$, and $x_4 = 98$.

12.28 Consider the following data from Exercise 12.13 on page 452.

y (wear)	x_1 (oil viscosity)	x_2 (load)
193	1.6	851
230	15.5	816
172	22.0	1058
91	43.0	1201
113	33.0	1357
125	40.0	1115

(a) Estimate σ^2 using multiple regression of y on x_1 and x_2 .

(b) Compute predicted values, a 95% confidence interval for mean wear, and a 95% prediction interval for observed wear if $x_1 = 20$ and $x_2 = 1000$.

12.29 Using the data from Exercise 12.28, test the following at the 0.05 level.

(a) $H_0: \beta_1 = 0$ versus $H_1: \beta_1 \neq 0$;

(b) $H_0: \beta_2 = 0$ versus $H_1: \beta_2 \neq 0$.

(c) Do you have any reason to believe that the model in Exercise 12.28 should be changed? Why or why not?

12.30 Use the data from Exercise 12.16 on page 453.

(a) Estimate σ^2 using the multiple regression of y on x_1 , x_2 , and x_3 ,

(b) Compute a 95% prediction interval for the observed gain with the three regressors at $x_1 = 15.0$, $x_2 = 220.0$, and $x_3 = 6.0$.

12.6 Choice of a Fitted Model through Hypothesis Testing

In many regression situations, individual coefficients are of importance to the experimenter. For example, in an economics application, $\beta_1, \beta_2, \dots$ might have some particular significance, and thus confidence intervals and tests of hypotheses on these parameters would be of interest to the economist. However, consider an industrial chemical situation in which the postulated model assumes that reaction yield is linearly dependent on reaction temperature and concentration of a certain catalyst. It is probably known that this is not the true model but an adequate approximation, so interest is likely to be not in the individual parameters but rather in the ability of the entire function to predict the true response in the range of the variables considered. Therefore, in this situation, one would put more emphasis on σ_Y^2 , confidence intervals on the mean response, and so forth, and likely deemphasize inferences on individual parameters.

The experimenter using regression analysis is also interested in deletion of variables when the situation dictates that, in addition to arriving at a workable prediction equation, he or she must find the “best regression” involving only variables that are useful predictors. There are a number of computer programs that sequentially arrive at the so-called best regression equation depending on certain criteria. We discuss this further in Section 12.9.

One criterion that is commonly used to illustrate the adequacy of a fitted regression model is the **coefficient of determination**, or R^2 .

Coefficient of
Determination, or
 R^2

$$R^2 = \frac{SSR}{SST} = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 1 - \frac{SSE}{SST}.$$

Note that this parallels the description of R^2 in Chapter 11. At this point the explanation might be clearer since we now focus on SSR as the **variability explained**. The quantity R^2 merely indicates what proportion of the total variation in the response Y is explained by the fitted model. Often an experimenter will report $R^2 \times 100\%$ and interpret the result as percent variation explained by the postulated model. The square root of R^2 is called the **multiple correlation coefficient** between Y and the set $x_1, x_2, \dots, x_k$. The value of R^2 for the case in Example 12.4, indicating the proportion of variation explained by the three independent variables x_1, x_2 , and x_3 , is

$$R^2 = \frac{SSR}{SST} = \frac{399.45}{438.13} = 0.9117,$$

which means that 91.17% of the variation in percent survival has been explained by the linear regression model.

The regression sum of squares can be used to give some indication concerning whether or not the model is an adequate explanation of the true situation. We can test the hypothesis H_0 that the **regression is not significant** by merely forming the ratio

$$f = \frac{SSR/k}{SSE/(n-k-1)} = \frac{SSR/k}{s^2}$$

and rejecting H_0 at the α -level of significance when $f > f_\alpha(k, n-k-1)$. For the data of Example 12.4, we obtain

$$f = \frac{399.45/3}{4.298} = 30.98.$$

From the printout in Figure 12.1, the P -value is less than 0.0001. This should not be misinterpreted. Although it does indicate that the regression explained by the model is significant, this does not rule out the following possibilities:

1. The linear regression model for this set of x 's is not the only model that can be used to explain the data; indeed, there may be other models with transformations on the x 's that give a larger value of the F -statistic.
2. The model might have been more effective with the inclusion of other variables in addition to x_1, x_2 , and x_3 or perhaps with the deletion of one or more of the variables in the model, say x_3 , which has a $P = 0.5916$.

The reader should recall the discussion in Section 11.5 regarding the pitfalls in the use of R^2 as a criterion for comparing competing models. These pitfalls are certainly relevant in multiple linear regression. In fact, in its employment in multiple regression, the dangers are even more pronounced since the temptation

to overfit is so great. One should always keep in mind that $R^2 \approx 1.0$ can always be achieved at the expense of error degrees of freedom when an excess of model terms is employed. However, $R^2 = 1$, describing a model with a near perfect fit, does not always result in a model that predicts well.

The Adjusted Coefficient of Determination (R_{adj}^2)

In Chapter 11, several figures displaying computer printout from both *SAS* and *MINITAB* featured a statistic called *adjusted R^2* or adjusted coefficient of determination. Adjusted R^2 is a variation on R^2 that provides an **adjustment for degrees of freedom**. The coefficient of determination as defined on page 407 cannot decrease as terms are added to the model. In other words, R^2 does not decrease as the error degrees of freedom $n - k - 1$ are reduced, the latter result being produced by an increase in k , the number of model terms. Adjusted R^2 is computed by dividing SSE and SST by their respective degrees of freedom as follows.

Adjusted R^2

$$R_{\text{adj}}^2 = 1 - \frac{SSE/(n - k - 1)}{SST/(n - 1)}.$$

To illustrate the use of R_{adj}^2 , Example 12.4 will be revisited.

How Are R^2 and R_{adj}^2 Affected by Removal of x_3 ?

The t -test (or corresponding F -test) for x_3 suggests that a simpler model involving only x_1 and x_2 may well be an improvement. In other words, the complete model with all the regressors may be an overfitted model. It is certainly of interest to investigate R^2 and R_{adj}^2 for both the full (x_1, x_2, x_3) and the reduced (x_1, x_2) models. We already know that $R_{\text{full}}^2 = 0.9117$ from Figure 12.1. The SSE for the reduced model is 40.01, and thus $R_{\text{reduced}}^2 = 1 - \frac{40.01}{438.13} = 0.9087$. Thus, more variability is explained with x_3 in the model. However, as we have indicated, this will occur even if the model is an overfitted model. Now, of course, R_{adj}^2 is designed to provide a statistic that punishes an overfitted model, so we might expect it to favor the reduced model. Indeed, for the full model

$$R_{\text{adj}}^2 = 1 - \frac{38.6764/9}{438.1308/12} = 1 - \frac{4.2974}{36.5109} = 0.8823,$$

whereas for the reduced model (deletion of x_3)

$$R_{\text{adj}}^2 = 1 - \frac{40.01/10}{438.1308/12} = 1 - \frac{4.001}{36.5109} = 0.8904.$$

Thus, R_{adj}^2 does indeed favor the reduced model and confirms the evidence produced by the t - and F -tests, suggesting that the reduced model is preferable to the model containing all three regressors. The reader may expect that other statistics would suggest rejection of the overfitted model. See Exercise 12.40 on page 471.

Test on an Individual Coefficient

The addition of any single variable to a regression system *will increase the regression sum of squares* and thus *reduce the error sum of squares*. Consequently, we must decide whether the increase in regression is sufficient to warrant using the variable in the model. As we might expect, the use of unimportant variables can reduce the effectiveness of the prediction equation by increasing the variance of the estimated response. We shall pursue this point further by considering the importance of x_3 in Example 12.4. Initially, we can test

$$H_0: \beta_3 = 0,$$

$$H_1: \beta_3 \neq 0$$

by using the t -distribution with 9 degrees of freedom. We have

$$t = \frac{b_3 - 0}{s\sqrt{c_{33}}} = \frac{-0.3433}{2.073\sqrt{0.0886}} = -0.556,$$

which indicates that β_3 does not differ significantly from zero, and hence we may very well feel justified in removing x_3 from the model. Suppose that we consider the regression of Y on the set (x_1, x_2) , the least squares normal equations now reducing to

$$\begin{bmatrix} 13.0 & 59.43 & 81.82 \\ 59.43 & 394.7255 & 360.6621 \\ 81.82 & 360.6621 & 576.7264 \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} 377.50 \\ 1877.5670 \\ 2246.6610 \end{bmatrix}.$$

The estimated regression coefficients for this reduced model are

$$b_0 = 36.094, \quad b_1 = 1.031, \quad b_2 = -1.870,$$

and the resulting regression sum of squares with 2 degrees of freedom is

$$R(\beta_1, \beta_2) = 398.12.$$

Here we use the notation $R(\beta_1, \beta_2)$ to indicate the regression sum of squares of the restricted model; it should not be confused with SSR , the regression sum of squares of the original model with 3 degrees of freedom. The new error sum of squares is then

$$SST - R(\beta_1, \beta_2) = 438.13 - 398.12 = 40.01,$$

and the resulting mean square error with 10 degrees of freedom becomes

$$s^2 = \frac{40.01}{10} = 4.001.$$

Does a Single Variable t -Test Have an F Counterpart?

From Example 12.4, the amount of variation in the percent survival that is attributed to x_3 , in the presence of the variables x_1 and x_2 , is

$$R(\beta_3 \mid \beta_1, \beta_2) = SSR - R(\beta_1, \beta_2) = 399.45 - 398.12 = 1.33,$$

which represents a small proportion of the entire regression variation. This amount of added regression is statistically insignificant, as indicated by our previous test on β_3 . An equivalent test involves the formation of the ratio

$$f = \frac{R(\beta_3 \mid \beta_1, \beta_2)}{s^2} = \frac{1.33}{4.298} = 0.309,$$

which is a value of the F -distribution with 1 and 9 degrees of freedom. Recall that the basic relationship between the t -distribution with v degrees of freedom and the F -distribution with 1 and v degrees of freedom is

$$t^2 = f(1, v),$$

and note that the f -value of 0.309 is indeed the square of the t -value of -0.56 .

To generalize the concepts above, we can assess the work of an independent variable x_i in the general multiple linear regression model

$$\mu_{Y \mid x_1, x_2, \dots, x_k} = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k$$

by observing the amount of regression attributed to x_i **over and above that attributed to the other variables**, that is, the regression on x_i *adjusted for the other variables*. For example, we say that x_1 is assessed by calculating

$$R(\beta_1 \mid \beta_2, \beta_3, \dots, \beta_k) = SSR - R(\beta_2, \beta_3, \dots, \beta_k),$$

where $R(\beta_2, \beta_3, \dots, \beta_k)$ is the regression sum of squares with $\beta_1 x_1$ removed from the model. To test the hypothesis

$$H_0: \beta_1 = 0,$$

$$H_1: \beta_1 \neq 0,$$

we compute

$$f = \frac{R(\beta_1 \mid \beta_2, \beta_3, \dots, \beta_k)}{s^2},$$

and compare it with $f_\alpha(1, n - k - 1)$.

Partial F -Tests on Subsets of Coefficients

In a similar manner, we can test for the significance of a *set* of the variables. For example, to investigate simultaneously the importance of including x_1 and x_2 in the model, we test the hypothesis

$$H_0: \beta_1 = \beta_2 = 0,$$

$$H_1: \beta_1 \text{ and } \beta_2 \text{ are not both zero,}$$

by computing

$$f = \frac{[R(\beta_1, \beta_2 \mid \beta_3, \beta_4, \dots, \beta_k)]/2}{s^2} = \frac{[SSR - R(\beta_3, \beta_4, \dots, \beta_k)]/2}{s^2}$$

and comparing it with $f_\alpha(2, n-k-1)$. The number of degrees of freedom associated with the numerator, in this case 2, equals the number of variables in the set being investigated.

Suppose we wish to test the hypothesis

$$H_0: \beta_2 = \beta_3 = 0,$$

$$H_1: \beta_2 \text{ and } \beta_3 \text{ are not both zero}$$

for Example 12.4. If we develop the regression model

$$y = \beta_0 + \beta_1 x_1 + \epsilon,$$

we can obtain $R(\beta_1) = SSR_{\text{reduced}} = 187.31179$. From Figure 12.1 on page 459, we have $s^2 = 4.29738$ for the full model. Hence, the f -value for testing the hypothesis is

$$\begin{aligned} f &= \frac{R(\beta_2, \beta_3 | \beta_1)/2}{s^2} = \frac{[R(\beta_1, \beta_2, \beta_3) - R(\beta_1)]/2}{s^2} = \frac{[SSR_{\text{full}} - SSR_{\text{reduced}}]/2}{s^2} \\ &= \frac{(399.45437 - 187.31179)/2}{4.29738} = 24.68278. \end{aligned}$$

This implies that β_2 and β_3 are not simultaneously zero. Using statistical software such as *SAS* one can directly obtain the above result with a P -value of 0.0002. Readers should note that in statistical software package output there are P -values associated with each individual model coefficient. The null hypothesis for each is that the coefficient is zero. However, it should be noted that the insignificance of any coefficient does not necessarily imply that it does not belong in the final model. It merely suggests that it is insignificant in the presence of all other variables in the problem. The case study at the end of this chapter illustrates this further.

12.7 Special Case of Orthogonality (Optional)

Prior to our original development of the general linear regression problem, the assumption was made that the independent variables are measured without error and are often controlled by the experimenter. Quite often they occur as a result of an *elaborately designed experiment*. In fact, we can increase the effectiveness of the resulting prediction equation with the use of a suitable experimental plan.

Suppose that we once again consider the $\mathbf{X}$ matrix as defined in Section 12.3. We can rewrite it as

$$\mathbf{X} = [\mathbf{1}, \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k],$$

where $\mathbf{1}$ represents a column of ones and $\mathbf{x}_j$ is a column vector representing the levels of x_j . If

$$\mathbf{x}'_p \mathbf{x}_q = \mathbf{0}, \quad \text{for } p \neq q,$$

the variables x_p and x_q are said to be *orthogonal* to each other. There are certain obvious advantages to having a completely orthogonal situation where $\mathbf{x}'_p \mathbf{x}_q = \mathbf{0}$

for all possible p and q , $p \neq q$, and, in addition,

$$\sum_{i=1}^n x_{ji} = 0, \quad j = 1, 2, \dots, k.$$

The resulting $\mathbf{X}'\mathbf{X}$ is a diagonal matrix, and the normal equations in Section 12.3 reduce to

$$nb_0 = \sum_{i=1}^n y_i, \quad b_1 \sum_{i=1}^n x_{1i}^2 = \sum_{i=1}^n x_{1i}y_i, \dots, b_k \sum_{i=1}^n x_{ki}^2 = \sum_{i=1}^n x_{ki}y_i.$$

An important advantage is that one is easily able to partition SSR into **single-degree-of-freedom components**, each of which corresponds to the amount of variation in Y accounted for by a given controlled variable. In the orthogonal situation, we can write

$$\begin{aligned} SSR &= \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \sum_{i=1}^n (b_0 + b_1x_{1i} + \dots + b_kx_{ki} - b_0)^2 \\ &= b_1^2 \sum_{i=1}^n x_{1i}^2 + b_2^2 \sum_{i=1}^n x_{2i}^2 + \dots + b_k^2 \sum_{i=1}^n x_{ki}^2 \\ &= R(\beta_1) + R(\beta_2) + \dots + R(\beta_k). \end{aligned}$$

The quantity $R(\beta_i)$ is the amount of the regression sum of squares associated with a model involving a single independent variable x_i .

To test simultaneously for the significance of a set of m variables in an orthogonal situation, the regression sum of squares becomes

$$R(\beta_1, \beta_2, \dots, \beta_m \mid \beta_{m+1}, \beta_{m+2}, \dots, \beta_k) = R(\beta_1) + R(\beta_2) + \dots + R(\beta_m),$$

and thus we have the further simplification

$$R(\beta_1 \mid \beta_2, \beta_3, \dots, \beta_k) = R(\beta_1)$$

when evaluating a single independent variable. Therefore, the contribution of a given variable or set of variables is essentially found by *ignoring* the other variables in the model. Independent evaluations of the worth of the individual variables are accomplished using analysis-of-variance techniques, as given in Table 12.4. The total variation in the response is partitioned into single-degree-of-freedom components plus the error term with $n - k - 1$ degrees of freedom. Each computed f -value is used to test one of the hypotheses

$$\left. \begin{array}{l} H_0: \beta_i = 0 \\ H_1: \beta_i \neq 0 \end{array} \right\} \quad i = 1, 2, \dots, k,$$

by comparing with the critical point $f_{\alpha}(1, n - k - 1)$ or merely interpreting the P -value computed from the f -distribution.

Table 12.4: Analysis of Variance for Orthogonal Variables

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	Computed f
β_1	$R(\beta_1) = b_1^2 \sum_{i=1}^n x_{1i}^2$	1	$R(\beta_1)$	$\frac{R(\beta_1)}{s^2}$
β_2	$R(\beta_2) = b_2^2 \sum_{i=1}^n x_{2i}^2$	1	$R(\beta_2)$	$\frac{R(\beta_2)}{s^2}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
β_k	$R(\beta_k) = b_k^2 \sum_{i=1}^n x_{ki}^2$	1	$R(\beta_k)$	$\frac{R(\beta_k)}{s^2}$
Error	SSE	$n - k - 1$	$s^2 = \frac{SSE}{n-k-1}$	
Total	$SST = S_{yy}$	$n - 1$		

Example 12.8: Suppose that a scientist takes experimental data on the radius of a propellant grain Y as a function of powder temperature x_1 , extrusion rate x_2 , and die temperature x_3 . Fit a linear regression model for predicting grain radius, and determine the effectiveness of each variable in the model. The data are given in Table 12.5.

Table 12.5: Data for Example 12.8

Grain Radius	Powder Temperature	Extrusion Rate	Die Temperature
82	150 (-1)	12 (-1)	220 (-1)
93	190 (+1)	12 (-1)	220 (-1)
114	150 (-1)	24 (+1)	220 (-1)
124	150 (-1)	12 (-1)	250 (+1)
111	190 (+1)	24 (+1)	220 (-1)
129	190 (+1)	12 (-1)	250 (+1)
157	150 (-1)	24 (+1)	250 (+1)
164	190 (+1)	24 (+1)	250 (+1)

Solution: Note that each variable is controlled at two levels, and the experiment is composed of the eight possible combinations. The data on the independent variables are coded for convenience by means of the following formulas:

$$x_1 = \frac{\text{powder temperature} - 170}{20},$$

$$x_2 = \frac{\text{extrusion rate} - 18}{6},$$

$$x_3 = \frac{\text{die temperature} - 235}{15}.$$

The resulting levels of x_1 , x_2 , and x_3 take on the values -1 and $+1$ as indicated in the table of data. This particular experimental design affords the orthogonal-

ity that we want to illustrate here. (A more thorough treatment of this type of experimental layout appears in Chapter 15.) The $\mathbf{X}$ matrix is

$$\mathbf{X} = \begin{bmatrix} 1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix},$$

and the orthogonality conditions are readily verified.

We can now compute coefficients

$$b_0 = \frac{1}{8} \sum_{i=1}^8 y_i = 121.75, \quad b_1 = \frac{1}{8} \sum_{i=1}^8 x_{1i} y_i = \frac{20}{8} = 2.5,$$

$$b_2 = \frac{\sum_{i=1}^8 x_{2i} y_i}{8} = \frac{118}{8} = 14.75, \quad b_3 = \frac{\sum_{i=1}^8 x_{3i} y_i}{8} = \frac{174}{8} = 21.75,$$

so in terms of the coded variables, the prediction equation is

$$\hat{y} = 121.75 + 2.5 x_1 + 14.75 x_2 + 21.75 x_3.$$

The analysis of variance in Table 12.6 shows independent contributions to SSR for each variable. The results, when compared to the $f_{0.05}(1, 4)$ critical point of 7.71, indicate that x_1 does not contribute significantly at the 0.05 level, whereas variables x_2 and x_3 are significant. In this example, the estimate for σ^2 is 23.1250. As for the single independent variable case, it should be pointed out that this estimate does not solely contain experimental error variation unless the postulated model is correct. Otherwise, the estimate is “contaminated” by lack of fit in addition to pure error, and the lack of fit can be separated out only if we obtain multiple experimental observations for the various (x_1, x_2, x_3) combinations.

Table 12.6: Analysis of Variance for Grain Radius Data

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Squares	Computed f	P -Value
β_1	$(2.5)^2(8) = 50.00$	1	50.00	2.16	0.2156
β_2	$(14.75)^2(8) = 1740.50$	1	1740.50	75.26	0.0010
β_3	$(21.75)^2(8) = 3784.50$	1	3784.50	163.65	0.0002
Error	92.50	4	23.13		
Total	5667.50	7			

Since x_1 is not significant, it can simply be eliminated from the model without altering the effects of the other variables. Note that x_2 and x_3 both impact the grain radius in a positive fashion, with x_3 being the more important factor based on the smallness of its P -value. ■

Exercises

12.31 Compute and interpret the coefficient of multiple determination for the variables of Exercise 12.1 on page 450.

12.32 Test whether the regression explained by the model in Exercise 12.1 on page 450 is significant at the 0.01 level of significance.

12.33 Test whether the regression explained by the model in Exercise 12.5 on page 450 is significant at the 0.01 level of significance.

12.34 For the model of Exercise 12.5 on page 450, test the hypothesis

$$H_0: \beta_1 = \beta_2 = 0,$$

$$H_1: \beta_1 \text{ and } \beta_2 \text{ are not both zero.}$$

12.35 Repeat Exercise 12.17 on page 461 using an F -statistic.

12.36 A small experiment was conducted to fit a multiple regression equation relating the yield y to temperature x_1 , reaction time x_2 , and concentration of one of the reactants x_3 . Two levels of each variable were chosen, and measurements corresponding to the coded independent variables were recorded as follows:

y	x_1	x_2	x_3
7.6	-1	-1	-1
8.4	1	-1	-1
9.2	-1	1	-1
10.3	-1	-1	1
9.8	1	1	-1
11.1	1	-1	1
10.2	-1	1	1
12.6	1	1	1

(a) Using the coded variables, estimate the multiple linear regression equation

$$\mu_{Y|x_1, x_2, x_3} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3.$$

(b) Partition SSR , the regression sum of squares, into three single-degree-of-freedom components attributable to x_1 , x_2 , and x_3 , respectively. Show an analysis-of-variance table, indicating significance tests on each variable.

12.37 Consider the electric power data of Exercise 12.5 on page 450. Test $H_0: \beta_1 = \beta_2 = 0$, making use of $R(\beta_1, \beta_2 | \beta_3, \beta_4)$. Give a P -value, and draw conclusions.

12.38 Consider the data for Exercise 12.36. Compute the following:

$$R(\beta_1 | \beta_0), \quad R(\beta_1 | \beta_0, \beta_2, \beta_3),$$

$$R(\beta_2 | \beta_0, \beta_1), \quad R(\beta_2 | \beta_0, \beta_1, \beta_3),$$

$$R(\beta_3 | \beta_0, \beta_1, \beta_2), \quad R(\beta_1, \beta_2 | \beta_3).$$

Comment.

12.39 Consider the data of Exercise 11.55 on page 437. Fit a regression model using weight and drive ratio as explanatory variables. Compare this model with the SLR (simple linear regression) model using weight alone. Use R^2 , R_{adj}^2 , and any t -statistics (or F -statistics) you may need to compare the SLR with the multiple regression model.

12.40 Consider Example 12.4. Figure 12.1 on page 459 displays a *SAS* printout of an analysis of the model containing variables x_1 , x_2 , and x_3 . Focus on the confidence interval of the mean response μ_Y at the (x_1, x_2, x_3) locations representing the 13 data points. Consider an item in the printout indicated by C.V. This is the **coefficient of variation**, which is defined by

$$\text{C.V.} = \frac{s}{\bar{y}} \cdot 100,$$

where $s = \sqrt{s^2}$ is the **root mean squared error**. The coefficient of variation is often used as yet another criterion for comparing competing models. It is a scale-free quantity which expresses the estimate of σ , namely s , as a percent of the average response $\bar{y}$. In competition for the “best” among a group of competing models, one strives for the model with a small value of C.V. Do a regression analysis of the data set shown in Example 12.4 but eliminate x_3 . Compare the full (x_1, x_2, x_3) model with the restricted (x_1, x_2) model and focus on two criteria: (i) C.V.; (ii) the widths of the confidence intervals on μ_Y . For the second criterion you may want to use the average width. Comment.

12.41 Consider Example 12.3 on page 447. Compare the two competing models.

$$\text{First order: } y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \epsilon_i,$$

$$\text{Second order: } y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$$

$$+ \beta_{11} x_{1i}^2 + \beta_{22} x_{2i}^2 + \beta_{12} x_{1i} x_{2i} + \epsilon_i.$$

Use R_{adj}^2 in your comparison. Test $H_0: \beta_{11} = \beta_{22} = \beta_{12} = 0$. In addition, use the C.V. discussed in Exercise 12.40.

12.42 In Example 12.8, a case is made for eliminating x_1 , powder temperature, from the model since the P -value based on the F -test is 0.2156 while P -values for x_2 and x_3 are near zero.

- (a) Reduce the model by eliminating x_1 , thereby producing a full and a restricted (or reduced) model, and compare them on the basis of R_{adj}^2 .
- (b) Compare the full and restricted models using the width of the 95% prediction intervals on a new observation. The better of the two models would be that with the tightened prediction intervals. Use the average of the width of the prediction intervals.

12.43 Consider the data of Exercise 12.13 on page 452. Can the response, wear, be explained adequately by a single variable (either viscosity or load) in an SLR rather than with the full two-variable regression? Justify your answer thoroughly through tests of hypotheses as well as comparison of the three competing models.

12.44 For the data set given in Exercise 12.16 on page 453, can the response be explained adequately by any two regressor variables? Discuss.

12.8 Categorical or Indicator Variables

An extremely important special-case application of multiple linear regression occurs when one or more of the regressor variables are **categorical, indicator, or dummy variables**. In a chemical process, the engineer may wish to model the process yield against regressors such as process temperature and reaction time. However, there is interest in using two different catalysts and somehow including “the catalyst” in the model. The catalyst effect cannot be measured on a continuum and is hence a categorical variable. An analyst may wish to model the price of homes against regressors that include square feet of living space x_1 , the land acreage x_2 , and age of the house x_3 . These regressors are clearly continuous in nature. However, it is clear that cost of homes may vary substantially from one area of the country to another. If data are collected on homes in the east, mid-west, south, and west, we have an indicator variable with **four categories**. In the chemical process example, if two catalysts are used, we have an indicator variable with two categories. In a biomedical example in which a drug is to be compared to a placebo, all subjects are evaluated on several continuous measurements such as age, blood pressure, and so on, as well as gender, which of course is categorical with two categories. So, included along with the continuous variables are two indicator variables: treatment with two categories (active drug and placebo) and gender with two categories (male and female).

Model with Categorical Variables

Let us use the chemical processing example to illustrate how indicator variables are involved in the model. Suppose y = yield and x_1 = temperature and x_2 = reaction time. Now let us denote the indicator variable by z . Let $z = 0$ for catalyst 1 and $z = 1$ for catalyst 2. The assignment of the (0, 1) indicator to the catalyst is arbitrary. As a result, the model becomes

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 z_i + \epsilon_i, \quad i = 1, 2, \dots, n.$$

Three Categories

The estimation of coefficients by the method of least squares continues to apply. In the case of three levels or categories of a single indicator variable, the model will

include **two** regressors, say z_1 and z_2 , where the $(0, 1)$ assignment is as follows:

$$\begin{matrix} & z_1 & z_2 \\ \begin{bmatrix} \mathbf{1} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} & & \begin{bmatrix} \mathbf{0} \\ \mathbf{1} \\ \mathbf{0} \end{bmatrix} \end{matrix},$$

where $\mathbf{0}$ and $\mathbf{1}$ are vectors of 0's and 1's, respectively. In other words, if there are ℓ categories, the model includes $\ell - 1$ actual model terms.

It may be instructive to look at a graphical representation of the model with three categories. For the sake of simplicity, let us assume a single continuous variable x . As a result, the model is given by

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_{1i} + \beta_3 z_{2i} + \epsilon_i.$$

Thus, Figure 12.2 reflects the nature of the model. The following are model expressions for the three categories.

$$E(Y) = (\beta_0 + \beta_2) + \beta_1 x, \quad \text{category 1,}$$

$$E(Y) = (\beta_0 + \beta_3) + \beta_1 x, \quad \text{category 2,}$$

$$E(Y) = \beta_0 + \beta_1 x, \quad \text{category 3.}$$

As a result, the model involving categorical variables essentially involves a **change in the intercept** as we change from one category to another. Here of course we are assuming that the **coefficients of continuous variables remain the same across the categories**.

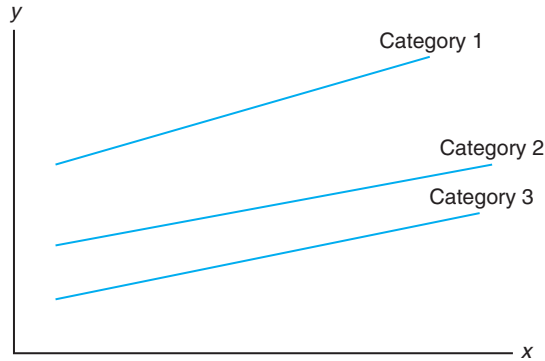


Figure 12.2: Case of three categories.

Example 12.9: Consider the data in Table 12.7. The response y is the amount of suspended solids in a coal cleansing system. The variable x is the pH of the system. Three different polymers are used in the system. Thus, “polymer” is categorical with three categories and hence produces two model terms. The model is given by

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_{1i} + \beta_3 z_{2i} + \epsilon_i, \quad i = 1, 2, \dots, 18.$$

Here we have

$$z_1 = \begin{cases} 1, & \text{for polymer 1,} \\ 0, & \text{otherwise,} \end{cases} \quad \text{and} \quad z_2 = \begin{cases} 1, & \text{for polymer 2,} \\ 0, & \text{otherwise.} \end{cases}$$

From the analysis in Figure 12.3, the following conclusions are drawn. The coefficient b_1 for pH is the estimate of the **common slope** that is assumed in the regression analysis. All model terms are statistically significant. Thus, pH and the nature of the polymer have an impact on the amount of cleansing. The signs and magnitudes of the coefficients of z_1 and z_2 indicate that polymer 1 is most effective (producing higher suspended solids) for cleansing, followed by polymer 2. Polymer 3 is least effective. 

Table 12.7: Data for Example 12.9

x , (pH)	y , (amount of suspended solids)	Polymer
6.5	292	1
6.9	329	1
7.8	352	1
8.4	378	1
8.8	392	1
9.2	410	1
6.7	198	2
6.9	227	2
7.5	277	2
7.9	297	2
8.7	364	2
9.2	375	2
6.5	167	3
7.0	225	3
7.2	247	3
7.6	268	3
8.7	288	3
9.2	342	3

Slope May Vary with Indicator Categories

In the discussion given here, we have assumed that the indicator variable model terms enter the model in an additive fashion. This suggests that the slopes, as in Figure 12.2, are constant across categories. Obviously, this is not always going to be the case. We can account for the possibility of varying slopes and indeed test for this condition of **parallelism** by including product or **interaction** terms between indicator terms and continuous variables. For example, suppose a model with one continuous regressor and an indicator variable with two levels is chosen. The model is given by

$$y = \beta_0 + \beta_1 x + \beta_2 z + \beta_3 xz + \epsilon.$$

		Sum of		Mean Square	F Value	Pr > F
Source	DF	Squares				
Model	3	80181.73127		26727.24376	73.68	<.0001
Error	14	5078.71318		362.76523		
Corrected Total	17	85260.44444				
R-Square	Coeff Var	Root MSE		y Mean		
0.940433	6.316049	19.04640		301.5556		
Parameter	Estimate	Error	t Value	Standard	Pr > t	
Intercept	-161.8973333	37.43315576	-4.32		0.0007	
x	54.2940260	4.75541126	11.42		<.0001	
z1	89.9980606	11.05228237	8.14		<.0001	
z2	27.1656970	11.01042883	2.47		0.0271	

Figure 12.3: SAS printout for Example 12.9.

This model suggests that for category 1 ($z = 1$),

$$E(y) = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)x,$$

while for category 2 ($z = 0$),

$$E(y) = \beta_0 + \beta_1x.$$

Thus, we allow for varying intercepts and slopes for the two categories. Figure 12.4 displays the regression lines with varying slopes for the two categories.

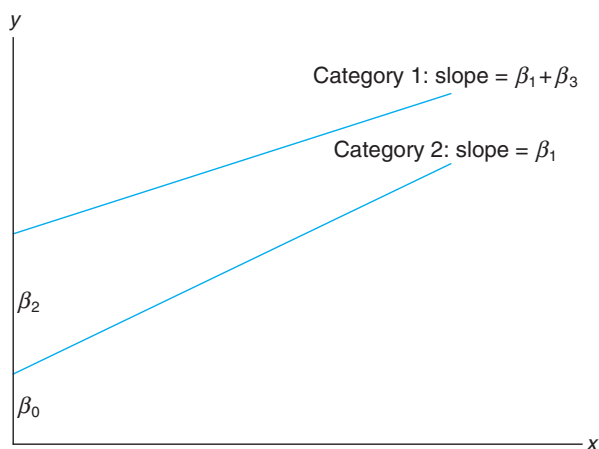


Figure 12.4: Nonparallelism in categorical variables.

In this case, β_0 , β_1 , and β_2 are positive while β_3 is negative with $|\beta_3| < \beta_1$. Obviously, if the interaction coefficient β_3 is insignificant, we are back to the common slope model.

Exercises

12.45 A study was done to assess the cost effectiveness of driving a four-door sedan instead of a van or an SUV (sports utility vehicle). The continuous variables are odometer reading and octane of the gasoline used. The response variable is miles per gallon. The data are presented here.

MPG	Car Type	Odometer	Octane
34.5	sedan	75,000	87.5
33.3	sedan	60,000	87.5
30.4	sedan	88,000	78.0
32.8	sedan	15,000	78.0
35.0	sedan	25,000	90.0
29.0	sedan	35,000	78.0
32.5	sedan	102,000	90.0
29.6	sedan	98,000	87.5
16.8	van	56,000	87.5
19.2	van	72,000	90.0
22.6	van	14,500	87.5
24.4	van	22,000	90.0
20.7	van	66,500	78.0
25.1	van	35,000	90.0
18.8	van	97,500	87.5
15.8	van	65,500	78.0
17.4	van	42,000	78.0
15.6	SUV	65,000	78.0
17.3	SUV	55,500	87.5
20.8	SUV	26,500	87.5
22.2	SUV	11,500	90.0
16.5	SUV	38,000	78.0
21.3	SUV	77,500	90.0
20.7	SUV	19,500	78.0
24.1	SUV	87,000	90.0

- (a) Fit a linear regression model including two indicator variables. Use (0, 0) to denote the four-door sedan.
- (b) Which type of vehicle appears to get the best gas mileage?

- (c) Discuss the difference between a van and an SUV in terms of gas mileage.

12.46 A study was done to determine whether the gender of the credit card holder was an important factor in generating profit for a certain credit card company. The variables considered were income, the number of family members, and the gender of the card holder. The data are as follows:

Profit	Income	Gender	Family Members
157	45,000	M	1
-181	55,000	M	2
-253	45,800	M	4
158	38,000	M	3
75	75,000	M	4
202	99,750	M	4
-451	28,000	M	1
146	39,000	M	2
89	54,350	M	1
-357	32,500	M	1
522	36,750	F	1
78	42,500	F	3
5	34,250	F	2
-177	36,750	F	3
123	24,500	F	2
251	27,500	F	1
-56	18,000	F	1
453	24,500	F	1
288	88,750	F	1
-104	19,750	F	2

- (a) Fit a linear regression model using the variables available. Based on the fitted model, would the company prefer male or female customers?
- (b) Would you say that income was an important factor in explaining the variability in profit?

12.9 Sequential Methods for Model Selection

At times, the significance tests outlined in Section 12.6 are quite adequate for determining which variables should be used in the final regression model. These tests are certainly effective if the experiment can be planned and the variables are orthogonal to each other. Even if the variables are not orthogonal, the individual t -tests can be of some use in many problems where the number of variables under investigation is small. However, there are many problems where it is necessary to use more elaborate techniques for screening variables, particularly when the experiment exhibits a substantial deviation from orthogonality. Useful measures of **multicollinearity** (linear dependency) among the independent variables are provided by the sample correlation coefficients $r_{x_i x_j}$. Since we are concerned only

with linear dependency among independent variables, no confusion will result if we drop the x 's from our notation and simply write $r_{x_i x_j} = r_{ij}$, where

$$r_{ij} = \frac{S_{ij}}{\sqrt{S_{ii}S_{jj}}}.$$

Note that the r_{ij} do not give true estimates of population correlation coefficients in the strict sense, since the x 's are actually not random variables in the context discussed here. Thus, the term *correlation*, although standard, is perhaps a misnomer.

When one or more of these sample correlation coefficients deviate substantially from zero, it can be quite difficult to find the most effective subset of variables for inclusion in our prediction equation. In fact, for some problems the multicollinearity will be so extreme that a suitable predictor cannot be found unless all possible subsets of the variables are investigated. Informative discussions of model selection in regression by Hocking (1976) are cited in the Bibliography. Procedures for detection of multicollinearity are discussed in the textbook by Myers (1990), also cited.

The user of multiple linear regression attempts to accomplish one of three objectives:

1. Obtain estimates of individual coefficients in a complete model.
2. Screen variables to determine which have a significant effect on the response.
3. Arrive at the most effective prediction equation.

In (1) it is known a priori that all variables are to be included in the model. In (2) prediction is secondary, while in (3) individual regression coefficients are not as important as the quality of the estimated response $\hat{y}$. For each of the situations above, multicollinearity in the experiment can have a profound effect on the success of the regression.

In this section, some standard sequential procedures for selecting variables are discussed. They are based on the notion that a single variable or a collection of variables should not appear in the estimating equation unless the variables result in a significant increase in the regression sum of squares or, equivalently, a significant increase in R^2 , the coefficient of multiple determination.

Illustration of Variable Screening in the Presence of Collinearity

Example 12.10: Consider the data of Table 12.8, where measurements were taken for nine infants. The purpose of the experiment was to arrive at a suitable estimating equation relating the length of an infant to all or a subset of the independent variables. The sample correlation coefficients, indicating the linear dependency among the independent variables, are displayed in the symmetric matrix

$$\begin{array}{c} \begin{array}{cccc} x_1 & x_2 & x_3 & x_4 \end{array} \\ \left[\begin{array}{cccc} 1.0000 & 0.9523 & 0.5340 & 0.3900 \\ 0.9523 & 1.0000 & 0.2626 & 0.1549 \\ 0.5340 & 0.2626 & 1.0000 & 0.7847 \\ 0.3900 & 0.1549 & 0.7847 & 1.0000 \end{array} \right] \end{array}$$

Table 12.8: Data Relating to Infant Length*

Infant Length, y (cm)	Age, x_1 (days)	Length at Birth, x_2 (cm)	Weight at Birth, x_3 (kg)	Chest Size at Birth, x_4 (cm)
57.5	78	48.2	2.75	29.5
52.8	69	45.5	2.15	26.3
61.3	77	46.3	4.41	32.2
67.0	88	49.0	5.52	36.5
53.5	67	43.0	3.21	27.2
62.7	80	48.0	4.32	27.7
56.2	74	48.0	2.31	28.3
68.5	94	53.0	4.30	30.3
69.2	102	58.0	3.71	28.7

*Data analyzed by the Statistical Consulting Center, Virginia Tech, Blacksburg, Virginia.

Note that there appears to be an appreciable amount of multicollinearity. Using the least squares technique outlined in Section 12.2, the estimated regression equation was fitted using the complete model and is

$$\hat{y} = 7.1475 + 0.1000x_1 + 0.7264x_2 + 3.0758x_3 - 0.0300x_4.$$

The value of s^2 with 4 degrees of freedom is 0.7414, and the value for the coefficient of determination for this model is found to be 0.9908. Regression sums of squares, measuring the variation attributed to each individual variable in the presence of the others, and the corresponding t -values are given in Table 12.9.

Table 12.9: t -Values for the Regression Data of Table 12.8

Variable x_1	Variable x_2	Variable x_3	Variable x_4
$R(\beta_1 \mid \beta_2, \beta_3, \beta_4)$	$R(\beta_2 \mid \beta_1, \beta_3, \beta_4)$	$R(\beta_3 \mid \beta_1, \beta_2, \beta_4)$	$R(\beta_4 \mid \beta_1, \beta_2, \beta_3)$
= 0.0644	= 0.6334	= 6.2523	= 0.0241
$t = 0.2947$	$t = 0.9243$	$t = 2.9040$	$t = -0.1805$

A two-tailed critical region with 4 degrees of freedom at the 0.05 level of significance is given by $|t| > 2.776$. Of the four computed t -values, **only variable x_3 appears to be significant**. However, recall that although the t -statistic described in Section 12.6 measures the worth of a variable adjusted for all other variables, it does not detect the potential importance of a variable in combination with a subset of the variables. For example, consider the model with only the variables x_2 and x_3 in the equation. The data analysis gives the regression function

$$\hat{y} = 2.1833 + 0.9576x_2 + 3.3253x_3,$$

with $R^2 = 0.9905$, certainly not a substantial reduction from $R^2 = 0.9907$ for the complete model. However, unless the performance characteristics of this particular combination had been observed, one would not be aware of its predictive potential. This, of course, lends support for a methodology that observes *all possible regressions* or a systematic sequential procedure designed to test subsets. 

Stepwise Regression

One standard procedure for searching for the “optimum subset” of variables in the absence of orthogonality is a technique called **stepwise regression**. It is based on the procedure of sequentially introducing the variables into the model one at a time. Given a predetermined size α , the description of the stepwise routine will be better understood if the methods of **forward selection** and **backward elimination** are described first.

Forward selection is based on the notion that variables should be inserted one at a time until a satisfactory regression equation is found. The procedure is as follows:

STEP 1. Choose the variable that gives the largest regression sum of squares when performing a simple linear regression with y or, equivalently, that which gives the largest value of R^2 . We shall call this initial variable x_1 . If x_1 is insignificant, the procedure is terminated.

STEP 2. Choose the variable that, when inserted in the model, gives the largest increase in R^2 , in the presence of x_1 , over the R^2 found in step 1. This, of course, is the variable x_j for which

$$R(\beta_j | \beta_1) = R(\beta_1, \beta_j) - R(\beta_1)$$

is largest. Let us call this variable x_2 . The regression model with x_1 and x_2 is then fitted and R^2 observed. If x_2 is insignificant, the procedure is terminated.

STEP 3. Choose the variable x_j that gives the largest value of

$$R(\beta_j | \beta_1, \beta_2) = R(\beta_1, \beta_2, \beta_j) - R(\beta_1, \beta_2),$$

again resulting in the largest increase of R^2 over that given in step 2. Calling this variable x_3 , we now have a regression model involving x_1 , x_2 , and x_3 . If x_3 is insignificant, the procedure is terminated.

This process is continued until the most recent variable inserted fails to induce a significant increase in the explained regression. Such an increase can be determined at each step by using the appropriate partial F -test or t -test. For example, in step 2 the value

$$f = \frac{R(\beta_2 | \beta_1)}{s^2}$$

can be determined to test the appropriateness of x_2 in the model. Here the value of s^2 is the mean square error for the model containing the variables x_1 and x_2 . Similarly, in step 3 the ratio

$$f = \frac{R(\beta_3 | \beta_1, \beta_2)}{s^2}$$

tests the appropriateness of x_3 in the model. Now, however, the value for s^2 is the mean square error for the model that contains the three variables x_1 , x_2 , and x_3 . If $f < f_\alpha(1, n - 3)$ at step 2, for a prechosen significance level, x_2 is not included

and the process is terminated, resulting in a simple linear equation relating y and x_1 . However, if $f > f_\alpha(1, n - 3)$, we proceed to step 3. Again, if $f < f_\alpha(1, n - 4)$ at step 3, x_3 is not included and the process is terminated with the appropriate regression equation containing the variables x_1 and x_2 .

Backward elimination involves the same concepts as forward selection except that one begins with all the variables in the model. Suppose, for example, that there are five variables under consideration. The steps are as follows:

STEP 1. Fit a regression equation with all five variables included in the model. Choose the variable that gives the smallest value of the regression sum of squares **adjusted for the others**. Suppose that this variable is x_2 . Remove x_2 from the model if

$$f = \frac{R(\beta_2 \mid \beta_1, \beta_3, \beta_4, \beta_5)}{s^2}$$

is insignificant.

STEP 2. Fit a regression equation using the remaining variables x_1 , x_3 , x_4 , and x_5 , and repeat step 1. Suppose that variable x_5 is chosen this time. Once again, if

$$f = \frac{R(\beta_5 \mid \beta_1, \beta_3, \beta_4)}{s^2}$$

is insignificant, the variable x_5 is removed from the model. At each step, the s^2 used in the F -test is the mean square error for the regression model at that stage.

This process is repeated until at some step the variable with the smallest adjusted regression sum of squares results in a significant f -value for some predetermined significance level.

Stepwise regression is accomplished with a slight but important modification of the forward selection procedure. The modification involves further testing at each stage to ensure the continued effectiveness of variables that had been inserted into the model at an earlier stage. This represents an improvement over forward selection, since it is quite possible that a variable entering the regression equation at an early stage might have been rendered unimportant or redundant because of relationships that exist between it and other variables entering at later stages. Therefore, at a stage in which a new variable has been entered into the regression equation through a significant increase in R^2 as determined by the F -test, all the variables already in the model are subjected to F -tests (or, equivalently, to t -tests) in light of this new variable and are deleted if they do not display a significant f -value. The procedure is continued until a stage is reached where no additional variables can be inserted or deleted. We illustrate the stepwise procedure in the following example.

Example 12.11: Using the techniques of stepwise regression, find an appropriate linear regression model for predicting the length of infants for the data of Table 12.8.

Solution: **STEP 1.** Considering each variable separately, four individual simple linear regression equations are fitted. The following pertinent regression sums of

squares are computed:

$$\begin{aligned} R(\beta_1) &= 288.1468, & R(\beta_2) &= 215.3013, \\ R(\beta_3) &= 186.1065, & R(\beta_4) &= 100.8594. \end{aligned}$$

Variable x_1 clearly gives the largest regression sum of squares. The mean square error for the equation involving only x_1 is $s^2 = 4.7276$, and since

$$f = \frac{R(\beta_1)}{s^2} = \frac{288.1468}{4.7276} = 60.9500,$$

which exceeds $f_{0.05}(1, 7) = 5.59$, the variable x_1 is significant and is entered into the model.

STEP 2. Three regression equations are fitted at this stage, all containing x_1 . The important results for the combinations (x_1, x_2) , (x_1, x_3) , and (x_1, x_4) are

$$R(\beta_2|\beta_1) = 23.8703, \quad R(\beta_3|\beta_1) = 29.3086, \quad R(\beta_4|\beta_1) = 13.8178.$$

Variable x_3 displays the largest regression sum of squares in the presence of x_1 . The regression involving x_1 and x_3 gives a new value of $s^2 = 0.6307$, and since

$$f = \frac{R(\beta_3|\beta_1)}{s^2} = \frac{29.3086}{0.6307} = 46.47,$$

which exceeds $f_{0.05}(1, 6) = 5.99$, the variable x_3 is significant and is included along with x_1 in the model. Now we must subject x_1 in the presence of x_3 to a significance test. We find that $R(\beta_1 | \beta_3) = 131.349$, and hence

$$f = \frac{R(\beta_1|\beta_3)}{s^2} = \frac{131.349}{0.6307} = 208.26,$$

which is highly significant. Therefore, x_1 is retained along with x_3 .

STEP 3. With x_1 and x_3 already in the model, we now require $R(\beta_2 | \beta_1, \beta_3)$ and $R(\beta_4 | \beta_1, \beta_3)$ in order to determine which, if any, of the remaining two variables is entered at this stage. From the regression analysis using x_2 along with x_1 and x_3 , we find $R(\beta_2 | \beta_1, \beta_3) = 0.7948$, and when x_4 is used along with x_1 and x_3 , we obtain $R(\beta_4 | \beta_1, \beta_3) = 0.1855$. The value of s^2 is 0.5979 for the (x_1, x_2, x_3) combination and 0.7198 for the (x_1, x_2, x_4) combination. Since neither f -value is significant at the $\alpha = 0.05$ level, the final regression model includes only the variables x_1 and x_3 . The estimating equation is found to be

$$\hat{y} = 20.1084 + 0.4136x_1 + 2.0253x_3,$$

and the coefficient of determination for this model is $R^2 = 0.9882$.

Although (x_1, x_3) is the combination chosen by stepwise regression, it is not necessarily the combination of two variables that gives the largest value of R^2 . In fact, we have already observed that the combination (x_2, x_3) gives $R^2 = 0.9905$. Of course, the stepwise procedure never observed this combination. A rational argument could be made that there is actually a negligible difference in performance

between these two estimating equations, at least in terms of percent variation explained. It is interesting to observe, however, that the backward elimination procedure gives the combination (x_2, x_3) in the final equation (see Exercise 12.49 on page 494). 

Summary

The main function of each of the procedures explained in this section is to expose the variables to a systematic methodology designed to ensure the eventual inclusion of the best combinations of the variables. Obviously, there is no assurance that this will happen in all problems, and, of course, it is possible that the multicollinearity is so extensive that one has no alternative but to resort to estimation procedures other than least squares. These estimation procedures are discussed in Myers (1990), listed in the Bibliography.

The sequential procedures discussed here represent three of many such methods that have been put forth in the literature and appear in various regression computer packages that are available. These methods are designed to be computationally efficient but, of course, do not give results for all possible subsets of the variables. As a result, the procedures are most effective for data sets that involve a **large number of variables**. For regression problems involving a relatively small number of variables, modern regression computer packages allow for the computation and summarization of quantitative information on all models for every possible subset of the variables. Illustrations are provided in Section 12.11.

Choice of P -Values

As one might expect, the choice of the final model with these procedures may depend dramatically on what P -value is chosen. In addition, a procedure is most successful when it is forced to test a large number of candidate variables. For this reason, any forward procedure will be most useful when a relatively large P -value is used. Thus, some software packages use a default P -value of 0.50.

12.10 Study of Residuals and Violation of Assumptions (Model Checking)

It was suggested earlier in this chapter that the residuals, or errors in the regression fit, often carry information that can be very informative to the data analyst. The $e_i = y_i - \hat{y}_i$, $i = 1, 2, \dots, n$, which are the numerical counterpart to the ϵ_i , the model errors, often shed light on the possible violation of assumptions or the presence of “suspect” data points. Suppose that we let the vector $\mathbf{x}_i$ denote the values of the regressor variables corresponding to the i th data point, supplemented by a 1 in the initial position. That is,

$$\mathbf{x}'_i = [1, x_{1i}, x_{2i}, \dots, x_{ki}].$$

Consider the quantity

$$h_{ii} = \mathbf{x}'_i(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i, \quad i = 1, 2, \dots, n.$$